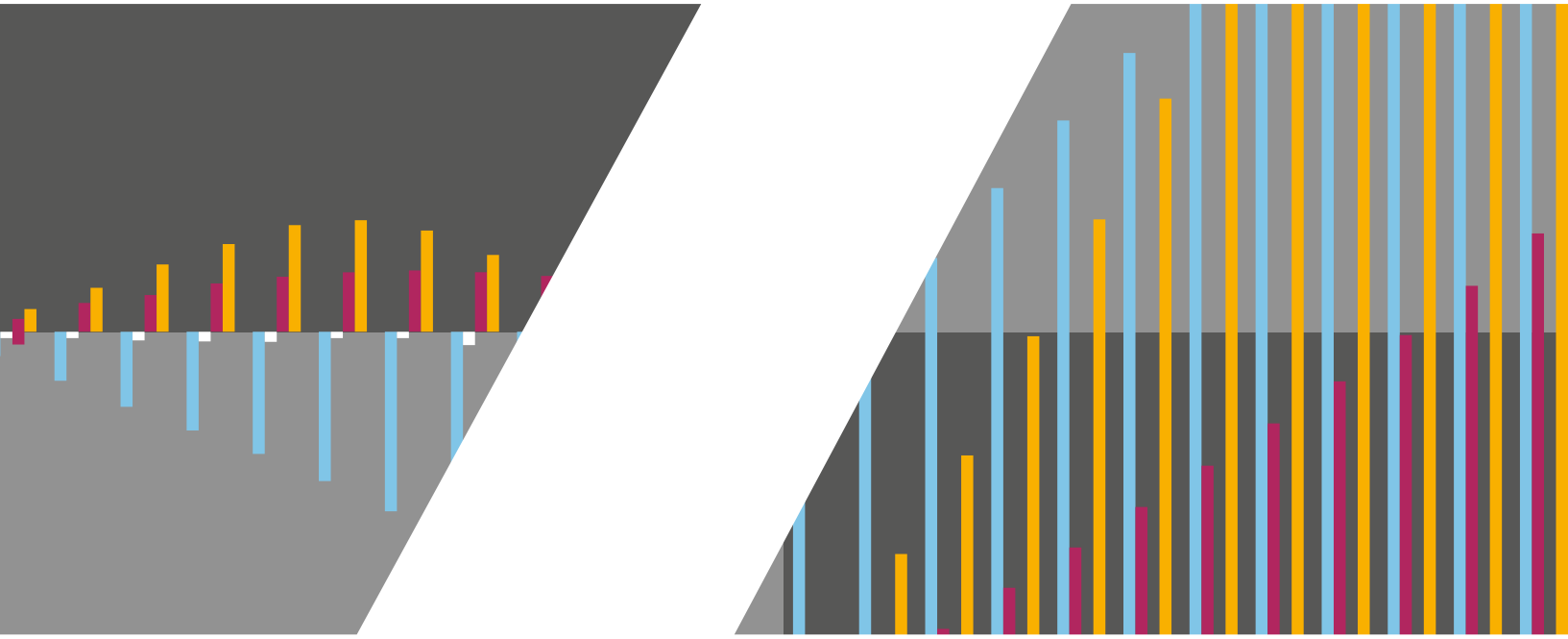
# HEALTH INEQUALITY MONITORING WORKBOOK



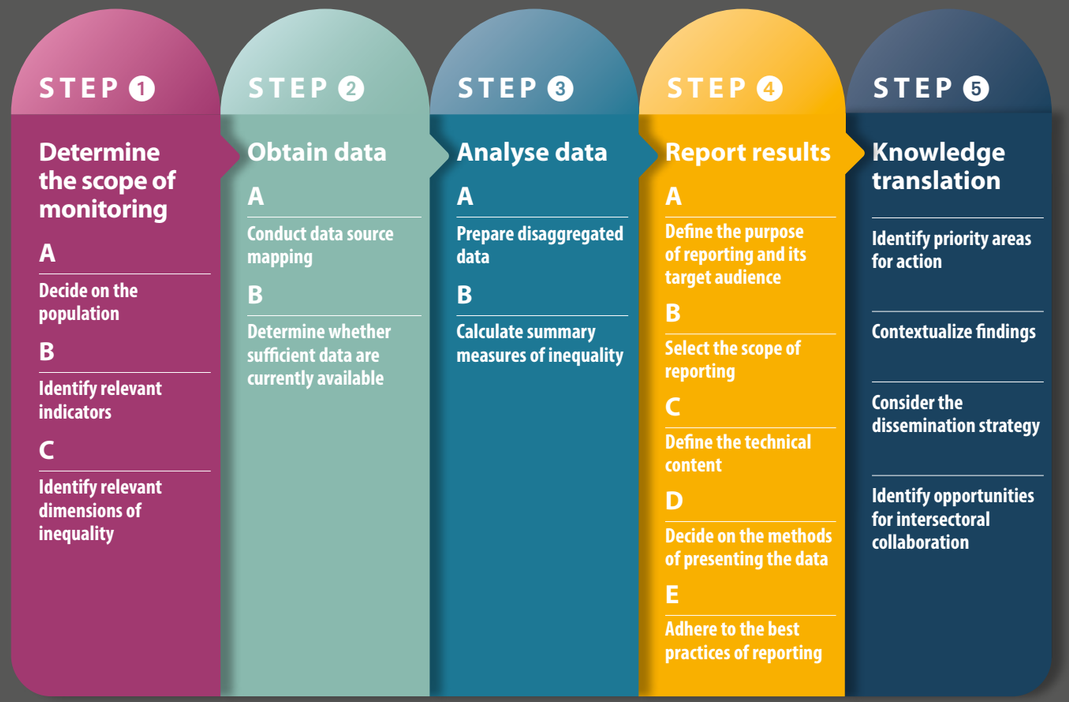
## EXERCISES TO GUIDE THE PROCESS OF HEALTH INEQUALITY MONITORING

SECOND EDITION

World Health Organization

## The cycle of health inequality monitoring




# HEALTH INEQUALITY MONITORING WORKBOOK

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SECOND EDITION

World Health Organization

WHO/DDI/DNA/MFI/2023.1

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# Contents

General considerations . . . 2

STEP 1 Determine the scope of monitoring . . . 4
- 1A. Decide on the population . . . 4
- 1B. Identify relevant indicators . . . 4
- 1C. Identify relevant dimensions of inequality . . . 6

STEP 2 Obtain data . . . 9
- 2A. Conduct data source mapping . . . 9
- 2B. Determine whether sufficient data are currently available . . . 12

STEP 3 Data analysis . . . 14
- 3A. Prepare disaggregated data . . . 15
- 3B. Calculate summary measures of inequality . . . 16

STEP 4 Report results . . . 19
- 4A. Define the purpose of reporting and its target audience . . . 20
- 4B. Select the scope of reporting . . . 21
- 4C. Define the technical content . . . 22
- 4D. Decide on the methods of presenting the data . . . 23
- 4E. Adhere to the best practices of reporting . . . 24

STEP 5 Knowledge translation . . . 25

This workbook contains exercises[1] that facilitate the application of the general concepts of health inequality monitoring in a specific context. Corresponding to the steps of the health inequality monitoring cycle, the workbook may be used to record personal reflections on key concepts, or to guide and document group planning sessions for inequality monitoring. Throughout the workbook there are prompts to make note of the key decisions made at each step of monitoring. This allows the workbook to serve as a detailed internal record of the monitoring process and of the rationale for how it was undertaken. Ideally, inequality monitoring should be repeated regularly; therefore, this workbook can serve as a reference for subsequent rounds of monitoring.

This workbook is intended to be used alongside the *National health inequality monitoring: a step-by-step manual* (), which provides further information on the health inequality monitoring process. Topic specific step-by-step manuals are available for *Immunization* () and *Sexual, reproductive, maternal, newborn, child and adolescent health* (). Furthermore, an overview of the health inequality monitoring process and its steps can be found here ().

[1] These exercises have been created for and used in health inequality monitoring training workshops and eLearning courses.

## General considerations

What health topic is the focus of this inequality monitoring exercise? (Max 20 characters)

Why is inequality monitoring being done? Why now?

Which policy cycles or windows of opportunity are important to keep in mind? (e.g. annual reporting cycles, programme or policy review cycles, meetings or conventions)

Which groups of people will use the knowledge generated from this monitoring exercise? (e.g. policy-makers, programme managers, affected communities or groups, representatives of non-profit organizations)





Which communities or groups are important to engage to ensure that the interests and needs of affected populations are represented?

How will the above communities or groups be involved in defining the scope of the monitoring, in interpreting the data and in disseminating results?

Which mechanisms will be in place to ensure the voices of disadvantaged groups are heard throughout the process?

# Step 1. Determine the scope of monitoring

## 1A. Decide on the population

What are the major strategies, programmes, policies and plans? Which populations do they serve? Which relevant populations do they not serve?

Which current or recent monitoring activities are related to , including community-led activities? Are there any inequality monitoring activities for in the specified topic ?

Which populations are underrepresented? Over-represented?

**DECISION:** Which population will your monitoring activity encompass? (Specify the geographical area and general characteristics of the affected population)



## 1B. Identify relevant indicators

How narrowly or broadly do you want to represent the topic of ? What is the rationale? How does this reflect the overall purpose and intended impact of monitoring, including the scope of practice of the desired knowledge users?

What are the key outcomes relevant to the population?

What are the key interventions relevant to the population?

Which indicator frameworks could be sources of valid indicators?

Which tracer indicators, if any, could be selected to represent the situation across a broader aspect of being monitored?

Are there composite indicators that could be considered?

**DECISION:** Which indicators are selected for inequality analysis? (List indicators)







## 1C. Identify relevant dimensions of inequality

Are the following common dimensions of inequality applicable to the monitoring population and package of indicators identified above? If so, which subgroups would be expected to be advantaged or disadvantaged?

| Dimension of inequality | Applicable? | How? |
|---|---|---|
| Sex | | |
| Economic status | | |
| Education level | | |
| Place of residence | | |
| Age | | |
| Subnational region | | |

Which other sources of disadvantage have you witnessed in the monitoring population that might have an impact on the topic(s) being monitored? How can these be reflected in context-specific dimensions of inequality?

**DECISION:** Which dimensions of inequality are relevant for monitoring inequality in ?







# Step 2. Obtain data

## 2A. Conduct data source mapping

Note: Data source templates for the five sheets in this step are available for download in spreadsheet format here ( ).

**Sheet 1. List data sources by type**

| Data source type | Name of data source | Year(s) of data collection | Notes (representativeness, population, data set accessibility etc.) |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

## Sheet 2. List data sources next to a unique data source number and indicate corresponding dimensions of inequality

| | | Dimensions of inequality | | | | | | |
|---|---|---|---|---|---|---|---|---|
| No. | Data source (year) | Sex | Economic status | Education level | Place of residence | Age | Subnational region | Other |
| | | | | | | | | |
| | | | | | | | | |
| | | | | | | | | |
| | | | | | | | | |
| | | | | | | | | |
| | | | | | | | | |
| | | | | | | | | |
| | | | | | | | | |

## Sheet 3. Indicators and corresponding unique data source numbers

| Indicator | Unique data source numbers (from Sheet 2) | Notes |
|---|---|---|
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |

**Sheet 4. Collate information about data sources (insert unique data source numbers from Sheet 2 in cells)**

| | Dimensions of inequality | | | | | | |
|---|---|---|---|---|---|---|---|
| **Indicator** | **Sex** | **Economic status** | **Education level** | **Place of residence** | **Age** | **Subnational region** | **Other** |
| | | | | | | | |
| | | | | | | | |
| | | | | | | | |
| | | | | | | | |
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| | | | | | | | |







## 2B. Determine whether sufficient data are currently available

Based on the results of the data source mapping, which data sources could be used for inequality monitoring?

What are the strengths and weaknesses of the candidate data sources? Consider the limitations inherent in the data collection protocol, the data quality issues, the accessibility and availability of data, the date of data collection (how recent was it?), and the existence of comparable data for benchmarking or assessing time trends.

Where are there gaps in data availability or quality? What could and should be done to address them?

Do data sources contain unique identifiers? If applicable, complete Sheet 5.

**Sheet 5. List common identifiers to assess the possibility of linking data**

| Common identifier | Unique data source numbers (from Sheet 2) |
|---|---|
| | |
| | |
| | |
| | |

**DECISION:** Which data sources will be used for monitoring?







## Step 3. Data analysis

Note: Exploration and comparison of health inequalities can be performed using the Health Equity Assessment Toolkit (HEAT and HEAT Plus) software applications (). HEAT allows users to explore disaggregated data within the WHO Health Inequality Data Repository through an online application, while HEAT Plus allow users to upload and work with their own inequality data.

### 3A. Prepare disaggregated data

How are indicators defined, including, if applicable, the numerators and denominators? Why were these definitions applied?

What are the interests and preferences of the affected population in terms of how subgroups are labelled and categorized? Consider each dimension of inequality identified in STEP 1C.

Do subgroups have small sample sizes? If so, how will this be addressed?

Which preparations are needed on the data set to proceed with analysis? Do data need to be cleaned? In the case of analysing household survey data, how will clustering, weighting and stratification be incorporated into the analysis?

**DECISION:** How are indicators and dimensions of inequality defined and categorized?

## 3B. Calculate summary measures of inequality

With which summary measures of inequality are you already familiar? What are their strengths and limitations? **Note: Common measures of inequality and their characteristics can be found here** (👣).

In which ways would you like to strengthen your capacity to use summary measures and conduct data analysis?







How will simple measures of inequality be applied to the analysis? In the table below, select absolute and relative simple measures of inequality, and indicate which two subgroups will be compared for each dimension of inequality.

| **Summary measure characteristic** | **Select relevant measures** | **Indicate two subgroups to be compared for each dimension of inequality** (e.g. economic status: wealth quintiles one and five) |
|---|---|---|
| Absolute | ☐ Difference | |
| Relative | ☐ Ratio | |

How will complex measures of inequality be applied to the analysis? In the table below, categorize each dimension of inequality as ordered or non-ordered, select absolute and relative complex measures of inequality, and indicate which two subgroups will be compared for each dimension of inequality.

| **Categorize** dimensions of inequality as ordered or non-ordered | Summary measure characteristics | | **Select** relevant measures[a] | **Indicate** the dimensions of inequality to which relevant measures will be applied |
|---|---|---|---|---|
| Ordered: | Weighted by population share | Absolute | ☐ Absolute concentration index | |
| | | | ☐ Slope index of inequality | |
| | | Relative | ☐ Relative concentration index | |
| | | | ☐ Relative index of inequality | |
| | Equal weighting | | (Not available) | |
| Non-ordered: | Weighted by population share | Absolute | ☐ Between-group standard deviation | |
| | | | ☐ Between-group variance | |
| | | | ☐ Mean difference from best-performing subgroup (weighted) | |
| | | | ☐ Mean difference from mean (weighted) | |
| | | Relative | ☐ Coefficient of variation | |
| | | | ☐ Index of disparity | |
| | | | ☐ Mean log deviation | |
| | | | ☐ Theil index | |
| | Equal weighting | Absolute | ☐ Mean difference from best-performing subgroup (unweighted) | |
| | | | ☐ Mean difference from mean (unweighted) | |
| | | Relative | ☐ Index of disparity (unweighted) | |
| Either ordered or non-ordered | Weighted by population share | Absolute | ☐ Population attributable risk | |
| | | Relative | ☐ Population attributable fraction | |

[a] This list includes common complex measures featured in the WHO Health Equity Assessment Toolkit (HEAT) and is not a comprehensive list of all summary measures of inequality. More information can be found in the **Summary Measures** section in HEAT's Technical Notes here ( ).

**DECISION:** Which summary measures of inequality will be calculated as part of monitoring?

# Step 4. Report results

Note that some of the questions for this step may need to be revisited once STEP 3 (data analysis) has been completed.

## 4A. Define the purpose of reporting and its target audience

Which goals and objectives do you hope to achieve through reporting?

Which audience(s) does reporting need to reach to achieve these goals and objectives?

What are the general characteristics of the audience(s)? Consider their skills (including reading level), prior knowledge, needs, priorities and interests.





**DECISION:** What is the purpose of reporting, and who is reporting targeted at?

## 4B. Select the scope of reporting

What are the most recent available data about the status of inequality in the specified topic?

If data are available for earlier time point(s), are they reasonably comparable with the most recent data? How would reporting time trends align with the purpose of reporting?

Which data are available for benchmarking across different populations, indicators or health topics? How would reporting benchmarking align with the purpose of reporting?

**DECISION:** Which data will be used to report the latest status of inequality? If applicable, how will trends over time and benchmarking be reported?

## 4C. Define the technical content

Which patterns can be identified in disaggregated data estimates for indicators across population subgroups?

Which main conclusions are supported by simple pairwise measures of inequality?

Do complex measures support these same conclusions? If not, where are the differences?

How much technical detail is warranted for the specified target audience?

**DECISION:** In addition to disaggregated data, which summary measures of inequality will be reported?

## 4D. Decide on the methods of presenting the data

Bearing in mind the decisions of STEPS 4A-C, what are the advantages and disadvantages of using the following tools to present data in support of the conclusions?

| Tool | Advantages | Disadvantages |
|---|---|---|
| Text | | |
| Tables | | |
| Figures | | |
| Maps | | |
| Interactive visuals | | |

Which other tools would be effective to present data?

**DECISION:** Which tools will be used to present the results?

## 4E. Adhere to the best practices of reporting

Which background information about policies, programmes and practices does the target audience need to know?

Which limitations are pertinent to the data sources, analysis and interpretation? How can they be communicated transparently?

Are there items on the checklist of best practices for reporting that cannot be fulfilled? If so, why?

**DECISION:** Which best practices of reporting are reflected in the reporting outputs?

## Step 5. Knowledge translation

Knowledge translation ensures the results of inequality monitoring are communicated to decision-makers and translated into policy and practice. Throughout this workbook, some initial knowledge translation considerations have been embedded in the previous steps of the inequality monitoring cycle.

See the *Inequality monitoring in Sexual, reproductive, maternal, newborn, child and adolescent health: a step-by-step manual* (📖) for an introduction to the topic of knowledge translation, which includes recommended resources for further reading.

Based on the findings of the inequality analysis, what are priorities for action?

What other forms of evidence (qualitative and quantitative) can help to contextualize the findings?

What are the key implications or recommendations for policy, programmes and practices? How can they best be communicated to the target audience?

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